CS6600: Computer Architecture

Assignment #2

Branch Predictor

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1 Introduction

This project involved the development of a generic branch predictor(in C++), to understand the behaviour of the actual branch predictors which are used in actual hardware. Using SPECint95 benchmarks containing GCC and JPEG traces a BiModal and GSHARE Branch Predictor have been designed.

2 Experiments

In this section, the address trace provided in the *gcc_trace.txt* and *jpeg_trace.txt* file was run with different values of PC lower order 'm' bits and 'n' bits for Global Branch History Register(GBHR). Variations in miss prediction rate for various values for 'm' and 'n' are being examined.

2.1 Bimodal Predictor

2.1.1 Plot #1

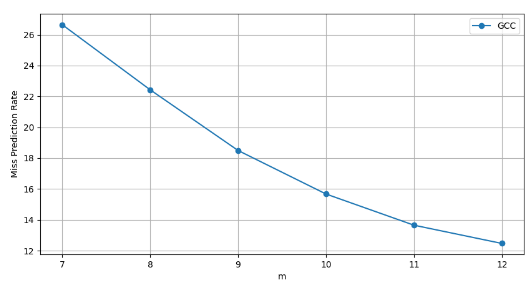

Figure 1: Miss Prediction Rate(%) vs m for GCC TRACE

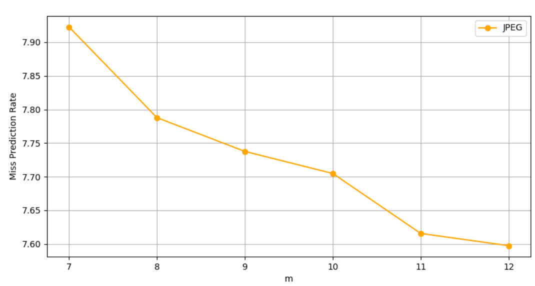

Figure 2: Miss Prediction Rate(%) vs m for JPEG TRACE

From the graphs we can observe that miss prediction rate decreases as 'm' increases for both GCC and JPEG traces. This is because, when 'm' is small, different branch might end up using the same counter leading to interference. More number of bits for indexing each branch will help in recognizing repetitive patterns and reduce the miss prediction rate.

From the plots, we can observe that for both GCC and JPEG the reduction in miss prediction rate is almost insignificant after 'm' is 11 bits. So, a bimodal predictor with 11 bit branch predictor table would be efficient for given traces.

2.2 Gshare Predictor

2.2.1 Plot #2

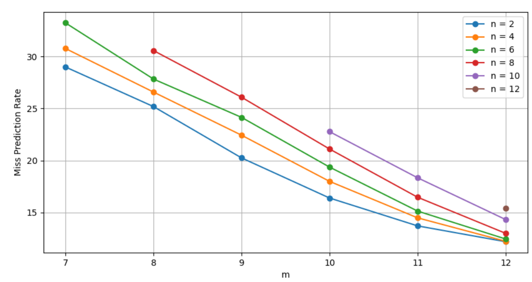

Figure 3: Miss Prediction Rate(%) vs m for different 'n' - GCC TRACE

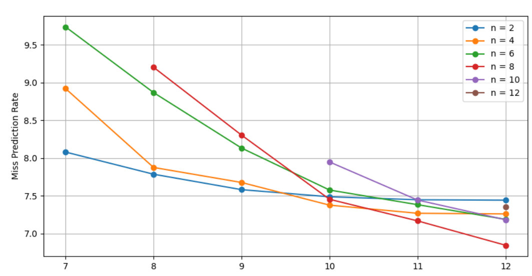

Figure 4: Miss Prediction Rate(%) vs m for different 'n' - JPEG TRACE

From the graphs we can observe that miss prediction rate decreases as 'm' increases for both GCC and JPEG traces like in the previous case for a given value of 'n'. But we can also see that miss prediction rate is higher for higher value of 'n' for a given 'm', which indicates that the number of bits used is more than that which is required for pattern recognition. And also miss prediction rate saturates quickly for lower values of 'n'.

From the plots, we can observe that for both GCC and JPEG the reduction in miss prediction rate is almost insignificant after 'm' is 11 bits. And as miss prediction rate increases with 'n', the lowest value of n(n = 2) will be preferred.
So, for efficient GSHARE branch predictor we can use m = 11 bits and n = 2 bits.